Registration Number :

VIT
Vellore Institute of Technology
CHENNAI

Continuous Assessment Test I - August 2024

Programme	B.Tech.(CSE)	Semester	Fall 2024-25
Course	Design and Analysis of Algorithms	Code	BCSE 204L
Faculty	Dr B Indira, Dr J Omana, Dr M Raja, Dr N Sivarakrishnan, Dr D Selvam, Dr G Kavipriya	Slot/Class No.	A2/CH2024250100959 /CH2024250101375 /CH2024250101371 /CH2024250102307 /CH2024250100963 /CH2024250101368
Time	90 Minutes	Max. Marks	50

Instructions:

- Answer all the FIVE questions.
- If any assumptions are required, assume the same and mention those assumptions in the answer script.
- Use of intelligence is highly appreciated.
- Your answer for all the questions should have both the 'design' component and the 'analysis component'
- The 'Design' component should consist: understanding of the problem, logic to develop the pseudocode, illustration, pseudocode.
- The 'Analysis' component should consist: Proof-of-Correctness, Computation of T(n), Time-complexity.

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1. Consider the following recursive algorithm:

Algorithm 1 $RecSum(n)$

1: **Input:** A positive integer $n > 0$
2: **Output:** Print n
3: **if** $n == 1$ **then**
4: return 1
5: **else**
6: return $(n + RecSum(n-1))$
7: **end if**

Understand the functionality of the above algorithm and answer the following.

(a) Trace the algorithm by assuming n = 5 and illustrates the description of the algorithm that you computes. [3 Marks]

(b) Determine the running time of the recursive algorithm RecSum(n). [2 Marks]

(c) Modify the algorithm to follow an iterative procedure to solve the same problem. [3 Marks]

(d) Compute the running time of your modified iterative algorithm and compare it with the recursive version. [2 Marks]

2. Consider two files X and Y of different sizes. Each file is represented by a sequence of characters where changes are often made in adjacent pairs of lines (e.g., function names or control structures). Your task is to find the **Longest Common Pair Subsequence (LCPS)** between these two sequences to identify common blocks of code. For the file **X = IFELSEFORIFRETURNN** and **Y = FORIFRETURNELSEIF** the **LCPS** identifies the common blocks of code between the above two files could be **FORIFRETURN**. **Design a pseudocode to solve the Longest Common Pair Subsequence (LCPS) problem which follows a dynamic programming strategy.** Your pseudocode should return the longest matching subsequence for the above given example. [10 marks]

[**Rubrics**: Logic for pseudocode: 2 marks, Illustration for pseudocode : 2 marks, Pseudocode : 3 marks, Proof-of-Correctness: 2 marks , Time-complexity :1 mark]

3. Let $A_1, A_2, \cdots A_n$ be n matrices, where A_i size is $p_{i-1} \times p_i$ for $i = 1, 2 \cdots n$. Let C be a cost matrix of size $n \times n$, where $C[i][j]$ is the minimum number of scalar multiplications to compute the chain $A_i \times A_{i+1} \times \cdots \times A_j$. Design a dynamic programming-based algorithm to compute the cost matrix C. [10 marks]

[**Rubrics**: Pseudocode: 4 marks, Illustration for pseudocode : 4 marks, Time-complexity :2 mark]

4. Given an array A of integers, and your task is to find a contiguous subarray that maximizes the sum. The subarray must have at least p positive elements and at least q negative elements. For example: $A = [4, -1, 2, 1, -5, 4, -3, 2, -1, 5, -6]$, $p = 2$ (minimum positive elements) , $q = 2$ (minimum negative elements). Design an algorithm to find the contiguous subarray that maximizes the sum $\sum_{i=j}^{k} A[i]$ and the algorithm should return the starting and ending indices of the subarray that meets the criteria, along with the maximum sum. [10 marks]

[**Rubrics**: Logic for pseudocode: 2 marks, Illustration: 3 marks, Pseudocode : 4 marks, Time-complexity :1 mark]

5. Consider an array A of n elements where each element may occur one or more times. For example: A[12, 45, 67, 12, 89, 67, 45, 12, 45, 89, 67, 45, 12, 89, 67, 89, 45], where
12 has occurred 4 times
45 has occurred 5 times
67 has occurred 4 times
89 has occurred 4 times
The expected output should be: [12, 67, 89, 45].
Design an algorithm to return the count of the number of occurrences of each element in the given set. Design a divide and conquer algorithm to arrange the elements in ascending order according to their count occurrence count. If two elements have the same count, they should be arranged in ascending order based on their value. Analyze the time complexity of the algorithm. [10 marks]

[**Rubrics**: Logic for pseudocode: 2 marks, Illustration for pseudocode : 2 marks, Pseudocode : 3 marks, Proof-of-Correctness: 2 marks , Time-complexity :1 mark]

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